

Grade 8 Mathematics

Մաթեմատիկա — 8-րդ դասարան

COURSE CODE	MTH-G8
PROGRAM	Middle School Mathematics · Միջին դպրոցի մաթեմատիկա
LEVEL	Grade 8 · typical age 13-14
DURATION	84-96 instructional hours (approx. 72 lessons)
PREREQUISITE	Grade 7 Mathematics or equivalent: rational number operations, proportional relationships, multi-step equations.
LEADS TO	Algebra I (MTH-A1)

COURSE DESCRIPTION

Դասընթացի նկարագրություն

Grade 8 is the bridge into formal algebra. The function arrives as an object — a rule assigning exactly one output to each input — and linear functions are studied through slope, intercepts, and rate of change. Students solve systems of two linear equations, work with integer exponents and scientific notation, meet irrational numbers through square roots, prove and apply the Pythagorean theorem, and study transformations as the basis of congruence and similarity. The year closes with bivariate data and the line of best fit.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

Grade 8 succeeds or fails on the concept of function. Վարժարան teachers introduce functions through multiple representations from the first lesson and insist on the definitional test — one output per input — rather than on the vertical line trick alone. Slope is taught as a rate of change with units attached, not as a memorized fraction of rise over run, because the rate interpretation is the one that transfers to Algebra I, to Calculus, and to every applied context. Transformations are taught with actual movement, because congruence defined as 'same shape and size' is a definition that cannot be used to prove anything.

PLACEMENT AND READINESS INDICATORS

Ընդունելություն և պատրաստվածություն

The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Recognizes proportional relationships and interprets the constant of proportionality	Grade 8 slope is Grade 7 proportionality generalized; without it, $y = mx + b$ is meaningless.
Solves multi-step equations with variables on both sides	Prerequisite for systems in Unit 3; a frequent gap.
Computes fluently with negative rational numbers	Slope calculations with negatives are the largest single source of Grade 8 errors.
Graphs points in all four quadrants	Prerequisite for every graphing task in the course.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

- 1 Know that numbers not rational are irrational, and approximate irrationals on a number line.
- 2 Apply properties of integer exponents to generate equivalent numerical expressions.
- 3 Use square and cube root symbols to represent and solve equations of the form $x^2 = p$ and $x^3 = p$.
- 4 Use scientific notation to express and compute with very large and very small quantities.
- 5 Graph proportional relationships and interpret the unit rate as the slope of the graph.
- 6 Derive $y = mx + b$ and explain why the graph of a linear equation is a straight line.
- 7 Solve linear equations in one variable, including those with no solution or infinitely many solutions.
- 8 Analyze and solve systems of two linear equations graphically, by substitution, and by elimination.
- 9 Understand a function as a rule assigning exactly one output to each input, and compare functions across representations.
- 10 Construct and interpret a function to model a linear relationship between two quantities.
- 11 Describe qualitatively the functional relationship shown by a graph.
- 12 Verify experimentally the properties of rotations, reflections, and translations.
- 13 Understand congruence and similarity through sequences of transformations.
- 14 Explain a proof of the Pythagorean theorem and apply it in two and three dimensions.
- 15 Know and use the volume formulas for cones, cylinders, and spheres.
- 16 Investigate patterns of association in bivariate data using scatter plots and two-way tables.

COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 • 10-11 LESSONS

Exponents, Roots, and the Real Numbers

Աստիճաններ, արմատներ և իրական թվեր

TOPICS COVERED

- Properties of integer exponents: product, quotient, power, and power of a product
- Zero and negative exponents and why they are defined as they are
- Simplifying expressions with integer exponents
- Square roots and cube roots
- Solving $x^2 = p$ and $x^3 = p$ and the two solutions of the square case
- Perfect squares and perfect cubes
- Rational and irrational numbers
- Why $\sqrt{2}$ is irrational (an accessible proof)
- Approximating an irrational number to a given precision
- Locating irrationals on the number line and comparing them
- Scientific notation for large and small numbers
- Operating with numbers in scientific notation
- Choosing units appropriate to a magnitude

LEARNING OBJECTIVES

- 1 Apply exponent properties to simplify an expression with integer exponents.
- 2 Explain why $a^0 = 1$ and why $a^{-n} = 1/a^n$, using the quotient property rather than a rule.
- 3 Approximate an irrational number to two decimal places and place it on a number line.
- 4 Multiply and divide numbers in scientific notation and interpret the result in context.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Simplifies integer-exponent expressions at 90% accuracy.
- Justifies the zero and negative exponent definitions from the quotient property.
- Computes in scientific notation correctly in 9 of 10 trials with correct final form.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
A negative exponent is believed to make the value negative.	It indicates a reciprocal, not a sign. Build the pattern $2^3, 2^2, 2^1, 2^0, 2^{-1}$ downward and let the student see the halving continue.
Exponents are added when bases are multiplied but bases are also multiplied: $2^3 \cdot 2^4 = 4^7$.	Expand once, fully, as repeated multiplication. The property should be a compression of something the student has written out.

MISCONCEPTION	RECOMMENDED RESPONSE
Every square root is assumed irrational, or $\sqrt{}$ is assumed to distribute over addition.	Test $\sqrt{9 + 16}$ against $\sqrt{9} + \sqrt{16}$ with a calculator. A single counterexample settles it permanently.

TEACHING NOTE

The proof that $\sqrt{2}$ is irrational is worth one full lesson even though no standard requires it. It is most students' first encounter with proof by contradiction, and it is memorable in a way that few Grade 8 lessons are.

UNIT 2 • 10-11 LESSONS**Linear Equations and Slope**

Գծային հավասարումներ և անկյունային գործակից

TOPICS COVERED

- Graphing proportional relationships and identifying the unit rate as slope
- Comparing two proportional relationships in different representations
- Slope as a rate of change with units
- Similar triangles and why slope is constant along a line
- Deriving $y = mx$ for a line through the origin and $y = mx + b$ generally
- Slope from two points, from a graph, from a table, and from an equation
- Positive, negative, zero, and undefined slope
- x- and y-intercepts and their meaning in context
- Solving linear equations with variables on both sides
- Equations with no solution and with infinitely many solutions
- Equations requiring expansion and collection of like terms
- Writing a linear equation from a context and interpreting its parameters

LEARNING OBJECTIVES

- 1 Compute slope from any representation and state its meaning in context with units.
- 2 Explain, using similar triangles, why the slope of a line is the same everywhere on it.
- 3 Solve linear equations including cases with no solution or infinitely many solutions, and interpret those cases.
- 4 Write an equation modelling a linear situation and interpret m and b in context.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Computes slope correctly from all four representations in 9 of 10 trials.
- Interprets slope and intercept in context with correct units in 9 of 10 trials.
- Solves linear equations at 90% accuracy, correctly identifying special cases.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Slope is computed as run over rise, or the subtraction order is inconsistent between numerator and denominator.	Label both points and write the formula with subscripts every time. Requiring the labels for two weeks eliminates most of these errors permanently.
A negative slope is confused with a line in a particular position rather than with a decreasing relationship.	Always describe slope in words first — 'for every one unit right, down two' — before writing the number.
An equation with no solution is answered as '$x = 0$'.	$0 = 5$ is a false statement, not a value of x . Contrast it with $0 = 0$ in the same lesson.

TEACHING NOTE

The similar-triangles argument for constant slope is the conceptual heart of the unit and is routinely skipped. It is what makes slope a property of the line rather than of the two points chosen, and it is required for anything later to make sense.

UNIT 3 • 8-9 LESSONS**Systems of Linear Equations**

Գծային հավասարումների համակարգեր

TOPICS COVERED

- What a solution to a system means
- Solving a system by graphing and reading the intersection
- The limits of graphical solutions
- Solving by substitution
- Solving by elimination
- Choosing an efficient method for a given system
- Systems with no solution: parallel lines
- Systems with infinitely many solutions: the same line
- Recognizing the special cases from the equations alone
- Writing a system to model a real situation
- Interpreting the solution in context

LEARNING OBJECTIVES

- 1 Explain that a solution to a system is a point satisfying both equations simultaneously.
- 2 Solve a system by graphing, substitution, and elimination and choose appropriately among them.
- 3 Determine from the equations alone whether a system has zero, one, or infinitely many solutions.
- 4 Write and solve a system modelling a real problem and interpret the answer in context.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves systems by all three methods at 90% accuracy.
- Correctly classifies the number of solutions from the equations in 9 of 10 trials.
- Models a real situation with a correct system in 8 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The solution is reported as a single number rather than an ordered pair.	A system in two variables has a two-part answer. Require both coordinates and a check in both equations.
In elimination, only one side of an equation is multiplied.	The entire equation must be scaled. Write the multiplier outside a bracket enclosing the whole equation.
Substitution is attempted without first isolating a variable.	Model the decision explicitly: which variable is cheapest to isolate, and why.

TEACHING NOTE

Systems are the first place where a student must choose a method rather than execute the only one available. Spend a lesson purely on method selection — presenting systems and asking only 'which method, and why?' without solving.

UNIT 4 • 9-10 LESSONS

Functions

Ֆունկցիաներ

TOPICS COVERED

- The definition of a function: exactly one output per input
- Domain and range informally
- Function notation and reading $f(x)$
- Functions as tables, graphs, equations, and verbal descriptions
- Testing whether a relation is a function in each representation
- Comparing two functions given in different representations
- Linear versus nonlinear functions and how to tell
- Constructing a linear function from two points or from a description
- Interpreting the rate of change and initial value of a linear function
- Describing a graph qualitatively: increasing, decreasing, constant, linear, nonlinear
- Sketching a graph from a verbal description of a situation

LEARNING OBJECTIVES

- 1 Determine whether a relation is a function from any representation and justify with the definition.
- 2 Interpret $f(3) = 7$ in the context of a described situation.
- 3 Compare the rates of change of two functions given in different forms.
- 4 Sketch a qualitative graph matching a verbal story and explain each feature.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Applies the function definition correctly in 9 of 10 trials across representations.
- Uses and interprets function notation correctly in 9 of 10 trials.
- Produces a qualitative graph consistent with a given story in 4 of 5 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
$f(x)$ is read as f times x.	Correct on first sight and every time thereafter. This misreading is fatal in Algebra II and beyond, and it is cheap to prevent now.
A relation is judged a function by the vertical line test alone, with no reference to the definition.	The test is a consequence of the definition, not a replacement. Require the definitional argument for tables and mappings, where no graph exists.
Any straight-looking graph is called linear without checking constant rate of change.	Check the table. Constant differences over equal intervals is the criterion.

TEACHING NOTE

Function notation is introduced here deliberately, ahead of many curricula. Students who become comfortable with $f(x)$ in Grade 8 have a substantially easier Algebra I, and the cost of introducing it early is low.

UNIT 5 • 9-10 LESSONS

Transformations, Congruence, and Similarity

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TOPICS COVERED

- Translations, reflections, and rotations as motions of the plane
- Properties preserved by rigid motions: length, angle measure, parallelism
- Describing a transformation with coordinates
- Composing transformations
- Congruence defined as the existence of a sequence of rigid motions
- Describing the sequence that maps one figure onto a congruent one
- Dilations and the scale factor
- Similarity defined via dilation and rigid motions
- Similar triangles and the angle-angle criterion
- The angle sum of a triangle established by a parallel-line argument
- The exterior angle theorem
- Angles formed by parallel lines cut by a transversal

LEARNING OBJECTIVES

- 1 Perform each rigid motion on the coordinate plane and state the coordinate rule.
- 2 Describe a specific sequence of transformations mapping one figure onto another.
- 3 Explain why the angle-angle criterion establishes similarity.
- 4 Prove informally that the angles of a triangle sum to 180° using parallel lines.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Performs and describes all three rigid motions accurately in 9 of 10 trials.

- Produces a valid transformation sequence for a congruence in 4 of 5 trials.
- Applies the AA criterion and parallel-line angle relationships at 85% accuracy.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Rotation direction and centre are ignored.	Every rotation requires a centre, an angle, and a direction. Require all three to be stated; a rotation without a centre is not a described transformation.
Congruence is defined as 'same shape and size', which cannot support an argument.	Define it as the existence of a sequence of rigid motions. This is the definition that makes proof possible in Geometry.
Similar figures are believed to have proportional angles as well as sides.	Angles are preserved exactly under dilation; only lengths scale. Measure to confirm.

TEACHING NOTE

This unit is the direct foundation of the transformational approach used in the Geometry course. The vocabulary and the habit of describing a specific mapping — not merely asserting congruence — should be established here.

UNIT 6 • 10-11 LESSONS

Pythagorean Theorem, Volume, and Bivariate Data

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TOPICS COVERED

- A proof of the Pythagorean theorem using area
- The converse and its use in testing for right angles
- Applying the theorem in two dimensions
- Applying the theorem in three dimensions
- Distance between two points on the coordinate plane
- Volume of cylinders, cones, and spheres
- Relationships among the three volume formulas
- Solving real problems with volume
- Scatter plots for bivariate measurement data
- Describing association: clustering, outliers, linearity, direction, strength
- Fitting a straight line informally and assessing the fit
- Interpreting the slope and intercept of a fitted line in context
- Two-way tables for bivariate categorical data
- Relative frequencies and evidence of association

LEARNING OBJECTIVES

- 1 Explain a proof of the Pythagorean theorem, not merely apply the formula.
- 2 Use the converse to determine whether a triangle is right-angled.

- 3 Compute the volume of a cone, cylinder, or sphere and explain the relationship among the formulas.
- 4 Fit a line to a scatter plot and interpret its slope and intercept in context.
- 5 Construct a two-way table and use relative frequencies to describe association.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Applies the theorem in two and three dimensions at 90% accuracy.
- Computes volumes of all three solids correctly with units in 9 of 10 trials.
- Interprets a fitted line's slope in context with correct units in 4 of 5 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The hypotenuse is not identified before the theorem is applied.	Mark the right angle and the side opposite it before writing anything. Most errors are identification errors, not computation errors.
The cone volume formula is applied without the factor of one third.	Demonstrate by filling a cone into a cylinder of the same base and height three times. The physical demonstration prevents the error.
Correlation is interpreted as causation.	Introduce a spurious-correlation example and require the student to propose an alternative explanation. This habit matters far beyond mathematics.

TEACHING NOTE

The distance formula should be derived here as the Pythagorean theorem applied to a coordinate grid, never presented as a separate formula to memorize. Students who see the derivation never forget it.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	Grade 7 proportionality and equation-solving; determines whether the student should instead begin Algebra I directly.
Unit Check	End of each unit · 45 min	Unit mastery criteria plus two items requiring justification or interpretation.
Fluency Probe	Every 3rd lesson · 5 min	Rational-number computation and slope calculation.
Function Concept Diagnostic	After Unit 4	Definition, notation, and multiple representations; the key predictor of Algebra I success.
Mid-Year Assessment	After Unit 3	Cumulative Units 1–3; weighted toward linear equations and slope.

ASSESSMENT	WHEN	WHAT IT MEASURES
End-of-Course Assessment	After Unit 6	All course outcomes; determines Algebra I placement and pacing.

Վարժարանը reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Standard	1 lesson/week, 60 min	Full course in approximately 50–55 weeks.
Accelerated	2 lessons/week	Full course in approximately 28–32 weeks.
Algebra I Bridge	2 lessons/week, 12 weeks	Units 2, 3, and 4 only — for students entering Algebra I with gaps in linear reasoning.
Intensive / Catch-Up	2–3 lessons/week	22–26 weeks with diagnostic-driven ordering.
Summer Program	3 lessons/week, 6 weeks	Linear functions intensive (Units 2 and 4).

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

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Forty to fifty minutes per lesson: eight to twelve problems including at least two multi-step applied problems and two cumulative review items. From Unit 2 onward, every problem involving slope or a linear model must include a sentence interpreting the answer in context with units — a bare number is marked incomplete. Students maintain a running error log of their own recurring mistakes, reviewed with the teacher at each unit boundary; this single practice measurably reduces repeated errors.

PROGRESS MILESTONES REPORTED TO PARENTS

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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- ◆ Applies integer exponent properties and justifies zero and negative exponents.
- ◆ Distinguishes rational from irrational numbers and approximates irrationals.
- ◆ Computes and interprets slope as a rate of change with units.
- ◆ Derives and uses $y = mx + b$ and explains why linear graphs are straight.
- ◆ Solves linear equations including no-solution and infinite-solution cases.
- ◆ Solves systems by graphing, substitution, and elimination and chooses appropriately.
- ◆ Understands the definition of a function and uses function notation correctly.
- ◆ Compares functions across tables, graphs, equations, and descriptions.
- ◆ Describes congruence and similarity through sequences of transformations.
- ◆ Explains a proof of the Pythagorean theorem and applies it in 2-D and 3-D.
- ◆ Finds volumes of cylinders, cones, and spheres.
- ◆ Fits and interprets a line on a scatter plot and reads a two-way table.
- ◆ Completed Grade 8 Mathematics — ready for Algebra I.

MATERIALS AND RESOURCES

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- Dynamic geometry software for the transformations unit (teacher-shared).
- Graphing tool for functions and systems; scientific calculator.
- Physical models of a cone and cylinder with matching base and height for the volume demonstration.
- Compass, protractor, and ruler.
- Վարժարան Grade 8 problem bank — approximately 1,500 items with a 250-item linear functions sub-bank.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.